Data-Driven Forecasting of Orbit-Averaged Thermospheric Density Using Ridge Regression, Random Forests, and Gradient-Boosted Trees

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Accurate prediction of thermospheric density is important for low-Earth-orbit drag modeling and orbit determination. This study evaluates three data-driven regressors—ridge regression, random forests, and LightGBM gradient-boosted trees—for predicting orbit-averaged density over a 72-hour horizon using the publicly released STORM-AI data set. Initial orbital conditions and summary statistics of 60 days of historical space-weather measurements are used to construct 25 predictors. Following data cleaning, 8,086 forecasting cases contribute approximately 3.41 million valid density observations. Cases are partitioned chronologically for training, validation, and testing. On 518,691 held-out test observations, ridge regression obtains a root-mean-square error of 8.72×10^{-14} kg/m^3, compared with 1.86×10^{-13} and 2.32×10^{-13} kg/m^3 for LightGBM and random forests, respectively. Ridge regression also trains more quickly, requiring 2.66 s versus 36.56 s and 283.54 s. The chronological partitions exhibit a pronounced density and solar-activity shift, which may contribute to the relative performance of the models. Results are compared with the published STORM-AI challenge findings while recognizing differences in evaluation data, metrics, and model inputs.

I. Introduction

Atmospheric drag is a major source of uncertainty in low-Earth-orbit (LEO) trajectory prediction. The drag acceleration depends on the local neutral atmospheric density, which varies with altitude, solar activity, geomagnetic forcing, and time. Errors in density prediction can therefore propagate into satellite position uncertainty and affect conjunction assessment and orbit maintenance. Predicting density over the next several days is challenging because the thermosphere responds to changing space-weather conditions.

The Satellite Tracking and Orbit Resilience Modeling using Artificial Intelligence (STORM-AI) challenge provides a public data set designed to evaluate machine-learning forecasts of orbit-averaged atmospheric density. The accompanying challenge analysis documents approaches ranging from gradient-boosted ensembles informed by empirical atmosphere models to sequence-based neural networks [1]. This project investigates a smaller, reproducible comparison using three standard regression methods rather than attempting to replicate the official challenge leaderboard.

The objective is to evaluate how model complexity affects predictive accuracy, computational cost, and temporal generalization. Specifically, ridge regression, random forest regression, and LightGBM are trained on the same engineered predictors and evaluated on a chronologically later test partition. A constant training-median predictor provides a simple reference. The principal finding is that the regularized linear model achieves the lowest test error, motivating an examination of distribution shift and the limitations of tree-based extrapolation.

II. Background

A. Atmospheric Density and Space Weather

To first order, the aerodynamic drag acceleration is proportional to atmospheric density,

$$\mathbf{a}_D = -\frac{1}{2}\frac{C_D A}{m}\rho \, \|\mathbf{v}_{\text{rel}}\|\mathbf{v}_{\text{rel}}, \tag{1}$$

where C_D is the drag coefficient, A/m is the area-to-mass ratio, ρ is neutral density, and $\mathbf{v}_{\text{rel}}$ is the velocity relative to the atmosphere. The density field is influenced by solar extreme-ultraviolet heating and geomagnetic energy deposition.

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Common proxies include the 10.7-cm solar radio flux ($F_{10.7}$), planetary geomagnetic indices (K_p and a_p), and the disturbance storm-time index (Dst).

B. STORM-AI and Prior Work

The STORM-AI challenge data include initial satellite states, historical space-weather observations, and 72-hour target sequences of orbit-averaged density sampled every 10 minutes [1, 2]. The published challenge report describes 8,118 training cases and separate hidden evaluation cases; the present study uses the publicly downloaded files after its own cleaning procedure. The winning approach employed LightGBM models with corrections relative to an empirical atmosphere model, whereas other highly ranked approaches incorporated temporal neural networks or additional physics-guided structure [1]. The reported winning-model RMSE improvements were 50.8% relative to JB2008 and 67.7% relative to MSIS on the challenge evaluation; during the May 2024 geomagnetic storm, the corresponding improvements were 6.1% and 34.7% [1]. These values motivate the present comparison but are not directly interchangeable with the raw RMSE reported here.

III. Technical Approach

A. Data Preparation and Feature Construction

The public `initial_states`, `omni2`, and `sat_density` archives are joined by forecasting-case identifier. GOES observations are not used. Each initial-state record supplies orbital and position information, while the corresponding OMNI2 file supplies a 60-day hourly history of space-weather conditions. The density file supplies the 72-hour forecast target. Invalid density values, missing-data sentinels, and implausible orbital states are removed. The resulting data contain 8,086 cases and 3,413,690 valid observation-level samples.

A total of 25 predictors are used: nine orbital/position variables, fifteen historical space-weather statistics, and forecast lead time. The weather features comprise the mean, standard deviation, and most recent valid value for each of $F_{10.7}$, K_p, Dst, a_p, and solar-wind plasma speed. The orbital variables include semimajor axis, eccentricity, inclination, right ascension of the ascending node, argument of perigee, true anomaly, latitude, longitude, and altitude. Forecast lead time ranges from 0 to 72 h. This approach compresses the input history into descriptive statistics rather than processing the full time series.

B. Temporal Partitioning

Forecasting cases are ordered chronologically and assigned approximately 70% to training, 15% to validation, and 15% to testing, with a 72-hour separation at partition boundaries. All observations from a case remain in the same partition. The separation reduces overlap between target horizons but does not eliminate overlap between the 60-day historical input windows. The held-out test partition contains 518,691 valid density observations; validation contains 515,950.

C. Regression Models

Atmospheric density spans multiple orders of magnitude, so the regressors are trained on the transformed target

$$y_i = \log_{10}\left(\rho_i / [1\ \mathrm{kg\,m^{-3}}]\right), \qquad \widehat{\rho}_i = 10^{\widehat{y}_i}\ \mathrm{kg\,m^{-3}}. \tag{2}$$

All reported errors are calculated after transforming predictions back into physical units.

Ridge regression minimizes a squared-error objective with L_2 regularization,

$$\widehat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \left\{ \sum_i (y_i - \beta_0 - \mathbf{x}_i^T \boldsymbol{\beta})^2 + \alpha \|\boldsymbol{\beta}\|_2^2 \right\}, \tag{3}$$

using standardized predictors and $\alpha = 10$. The random forest uses 100 trees, maximum depth 18, minimum leaf size five, and 80% of predictors considered at each split. LightGBM uses 500 boosting iterations, learning rate 0.05, 31 leaves, and minimum child size 30. The implementations use scikit-learn and LightGBM with fixed random seed 42. No hyperparameter optimization or storm-specific retraining is performed.

D. Evaluation Metrics

The principal metrics are root-mean-square error (RMSE) and mean absolute error (MAE),

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(\widehat{\rho}_i - \rho_i)^2}, \tag{4}$$

$$\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N}|\widehat{\rho}_i - \rho_i|. \tag{5}$$

Mean absolute percentage error (MAPE) is also reported as $100N^{-1}\sum_i |\widehat{\rho}_i - \rho_i|/|\rho_i|$. Because MAPE weights errors heavily when true density is small, it is interpreted alongside absolute errors. A constant baseline predicts the training-set median density, 1.083973×10^{-12} kg/m^3, for every observation. Training times are wall-clock measurements on the same local machine; they exclude preprocessing and are not hardware-normalized against published studies.

IV. Results

A. Overall Prediction Accuracy

Table 1 summarizes held-out test results. Ridge regression produces the smallest RMSE, MAE, and MAPE of the three learned models. Its test RMSE is 53.1% lower than LightGBM and 62.5% lower than the random forest. LightGBM reduces RMSE by 20.1% relative to the random forest. Relative to the constant-median baseline, ridge regression reduces test RMSE by approximately 91.4%; the baseline result should be interpreted as a simple reference rather than a physical atmospheric-model benchmark.

Table 1 Held-out test performance (518,691 observations for trained models).

Model	RMSE (kg/m^3)	MAE (kg/m^3)	MAPE (%)	Train (s)
Median baseline	1.013×10^{-12}	1.012×10^{-12}	1921.03	—
Random forest	2.324×10^{-13}	1.244×10^{-13}	144.50	283.54
LightGBM	1.857×10^{-13}	1.142×10^{-13}	138.06	36.56
Ridge	8.717×10^{-14}	5.245×10^{-14}	67.42	2.66

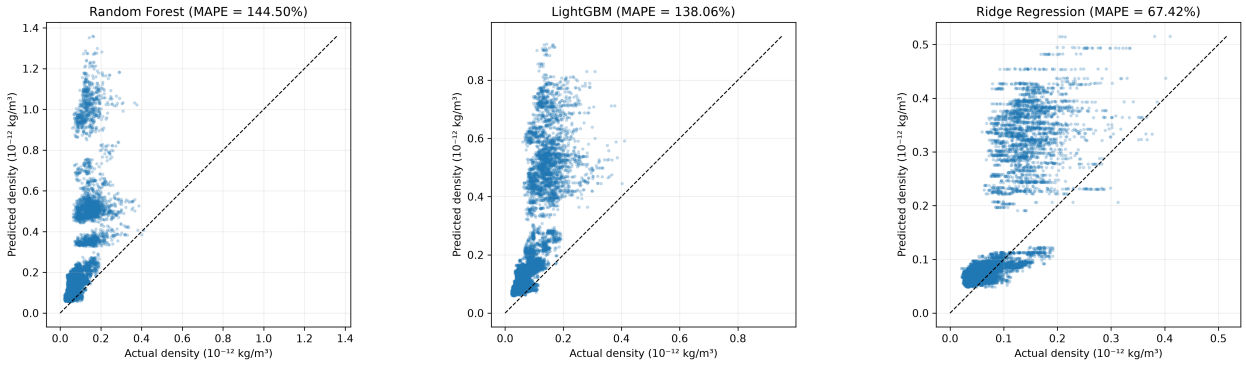


Fig. 1 Predicted versus observed density on the chronological test partition for the three regression models.

Figure 1 further illustrates differences between the learned models. The random forest and LightGBM predictions exhibit substantial positive bias for many low-density test observations, whereas ridge regression produces predictions closer to the observed density scale. This behavior is consistent with the distribution shift between the training and test partitions. However, the scatter plots alone do not establish that extrapolation limitations are the sole cause of the observed differences.

The computational difference is also substantial. Ridge training is approximately 13.8 times faster than LightGBM and 106 times faster than the random forest under the recorded local conditions. These differences demonstrate that, for this feature representation and split, increasing model flexibility did not improve held-out accuracy.

B. Distribution Shift and Forecast Horizon

The chronological partitions differ considerably in target scale. The training-period mean density is approximately 1.51×10^{-12} kg/m^3, compared with 4.59×10^{-13} kg/m^3 for validation and 7.24×10^{-14} kg/m^3 for testing. Mean historical $F_{10.7}$ also decreases from approximately 107.6 in training to 79.4 in validation and 70.9 in testing. Median altitude increases from approximately 456.5 km in training to 501.8 km in testing. These differences suggest that the test set represents a lower-density regime, due to a combination of solar conditions and orbital characteristics.

Decision-tree ensembles partition the predictor space and aggregate values learned from the training sample. Consequently, their predictions may be less capable of extrapolating to a substantially different target regime than a log-linear model with an appropriate directional relationship. This is a plausible explanation for the strong ridge results, not a demonstrated causal mechanism; further controlled tests would be needed to isolate the effects of altitude, solar activity, and chronology.

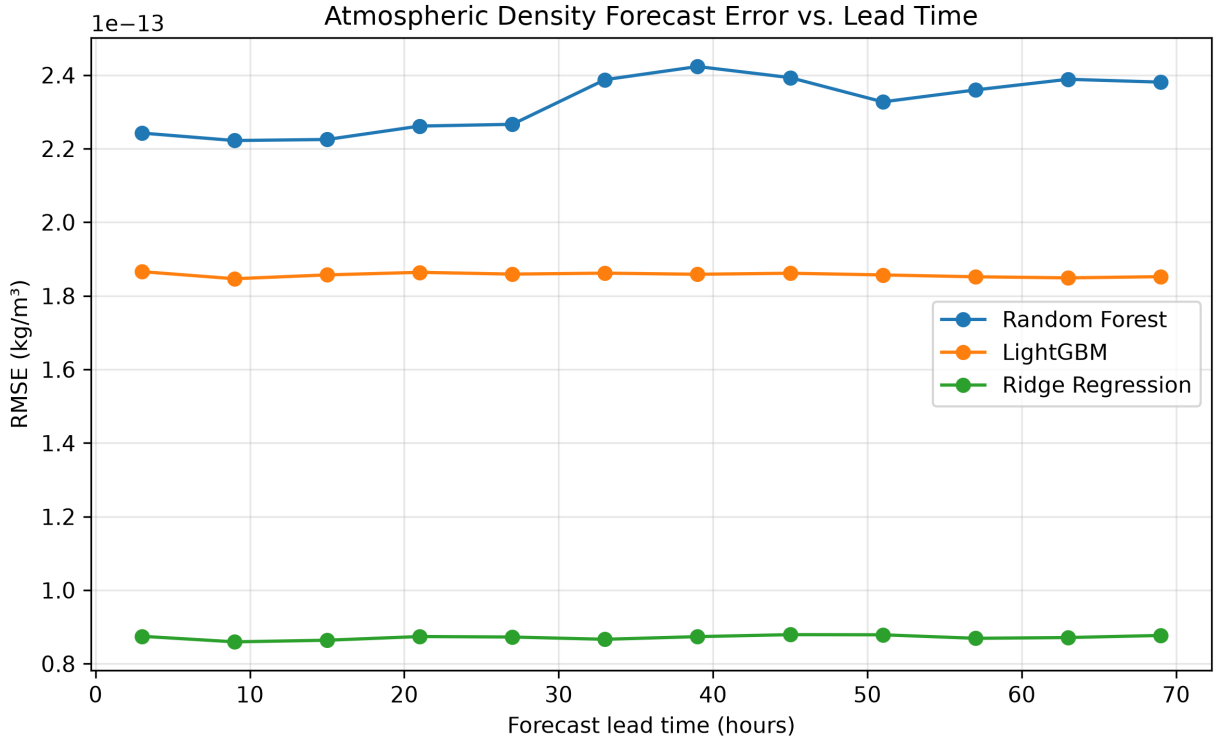


Fig. 2 Test RMSE versus forecast lead time, aggregated into six-hour intervals.

The lead-time analysis indicates relatively modest changes in error across the 72-hour window for the learned models. Since orbital conditions and weather summary features are fixed within each case, and only lead time varies across target rows, the feature representation may not capture rapid changes in thermospheric forcing during the forecast. More detailed temporal inputs would be required to test that hypothesis.

C. Feature Interpretation

The ridge model assigns a large negative standardized coefficient to semimajor axis and positive coefficients to solar-flux statistics, broadly consistent with decreasing atmospheric density at higher orbital altitudes and increased thermospheric heating during greater solar activity. Geomagnetic features also contribute. Because several orbital and weather predictors are correlated, coefficient magnitude alone does not establish physical causality or unique feature

importance.

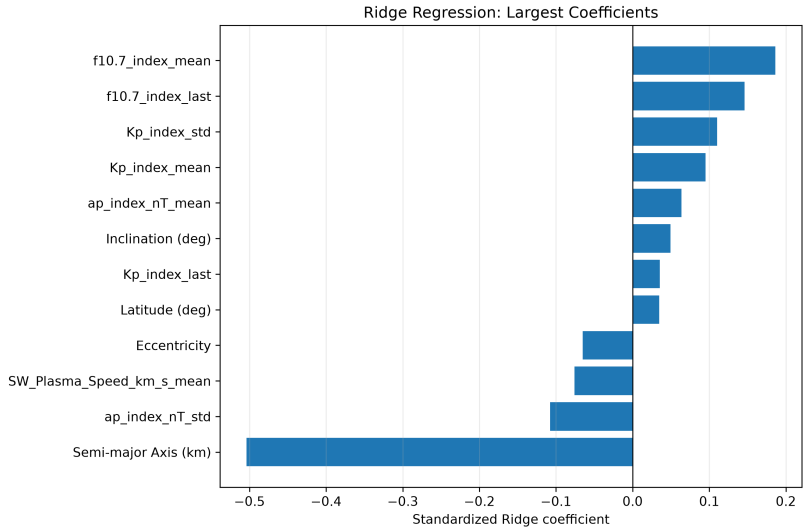


Fig. 3 Largest-magnitude standardized ridge regression coefficients. Coefficients describe associations in the fitted model and should not be interpreted as causal effects.

D. Comparison with Published STORM-AI Results

Table 2 places the current study alongside quantitative findings from the STORM-AI challenge report [1]. The official winning model improved RMSE relative to JB2008 and MSIS by 50.8% and 67.7%, respectively, whereas the present ridge model improves RMSE relative to a constant training-median baseline by approximately 91.4%. These percentages have *different denominators, test cases, and evaluation procedures* and cannot be ranked against one another. In particular, the competition's empirical-model baselines are substantially more informative than the constant baseline used here.

Table 2 Quantitative comparisons under distinct evaluation protocols.

This study: chronological public-data test	Published STORM-AI challenge [1]
Ridge vs. LightGBM: 53.1% lower RMSE	Winning model vs. JB2008: 50.8% lower RMSE
Ridge vs. random forest: 62.5% lower RMSE	Winning model vs. MSIS: 67.7% lower RMSE
Ridge vs. constant median: 91.4% lower RMSE	May 2024 storm: 6.1% lower RMSE vs. JB2008 and 34.7% vs. MSIS

The methodological distinction is equally important. The challenge-winning Bimasakti approach employed an ensemble of LightGBM models to learn corrections relative to an empirical atmospheric-density model, NRLMSIS 2.1 [1]. By contrast, the LightGBM model developed here predicts logarithmic density directly from orbital conditions and summary statistics of historical space weather. Consequently, the two implementations differ substantially despite using the same underlying gradient-boosting framework.

The published competition also demonstrated the difficulty of forecasting density during strong geomagnetic disturbances. Although the winning approach reduced RMSE by 50.8% relative to JB2008 and 67.7% relative to MSIS in the reported overall comparison, these improvements decreased to 6.1% and 34.7%, respectively, during the May 2024 geomagnetic storm [1]. This reduction illustrates that performance under extreme space-weather conditions can differ considerably from aggregate performance.

Other highly ranked approaches incorporated more detailed temporal representations. The Millennial-IUP submission combined convolutional and recurrent neural-network components, including LSTM and GRU architectures, whereas the Digantara approach incorporated gradient-boosted prediction with a temporal-learning architecture [1]. These

methods can represent temporal relationships that are discarded when a 60-day weather history is reduced to means, standard deviations, and most recent values.

Finally, the official challenge evaluated predictions on separate hidden cases using a time-weighted metric normalized relative to MSIS. The present study instead reports unweighted physical-unit RMSE on its own chronological partition. The results therefore support comparisons among the three implemented regressors, but do not establish superiority over the published STORM-AI submissions.

V. Conclusions

Three regression approaches were evaluated for 72-hour orbit-averaged thermospheric-density prediction using the publicly available STORM-AI data set. With 25 engineered predictors and a chronological test partition, ridge regression achieved an RMSE of 8.72×10^{-14} kg/m^3, outperforming LightGBM and random forest by 53.1% and 62.5%, respectively. It also trained much faster. These results demonstrate that a relatively simple regularized model can be competitive when the target is transformed logarithmically and the available predictors capture dominant altitude and solar-activity relationships.

The result is subject to important limitations. The chronological partitions differ substantially in density, altitude, and solar activity; 60-day historical input windows can overlap across split boundaries; and the engineered weather summaries do not encode the full temporal evolution of geomagnetic disturbances. The present baseline is a constant predictor, not JB2008 or MSIS, and the results are not directly comparable to official STORM-AI leaderboard scores. Future work should assess performance by geomagnetic-storm intensity, incorporate physics-based density priors and full time-series weather inputs, and evaluate on a benchmark split matching the published challenge protocol.

References

[1] S. Sanchez-Hurtado *et al.*, "AI Challenge for Satellite Tracking and Orbit Resilience Modeling (STORM-AI): Data Set, Design, and Results," *Space Weather*, Vol. 24, 2026, e2026SW004994. https://doi.org/10.1029/2026SW004994.

[2] MIT ARCLab, "STORM-AI Data Set," Harvard Dataverse, 2025. https://doi.org/10.7910/DVN/U6K6MJ.

[3] MIT ARCLab, "STORM-AI Devkit 2025," software repository. https://github.com/ARCLab-MIT/STORM-AI-devkit-2025.